Centrally Symmetric Floating Bodies and Inscribed Hexagons

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The Floating Body Problem

Scottish Book Problem 19, S. Ulam

Is a solid of uniform density which floats in water in every position necessarily a sphere?

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In 2D, we study the cross-section of a long cylinder with relative density ρ.

- The waterline cuts off a fixed area fraction determined by ρ.
- The induced boundary ratio is the *perimetral density* σ.

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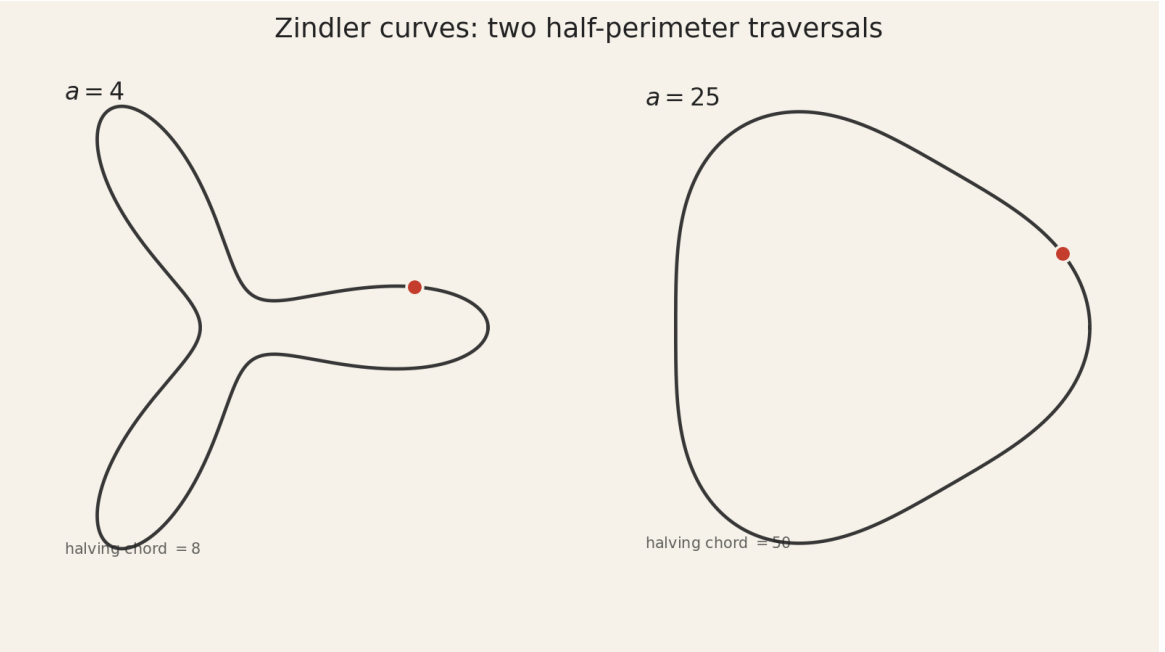
Floating bodies and geometry

- **Archimedes:** law of buoyancy and early stability results for floating solids.
- **P. Bouguer (1746):** introduced the *metacenter*, the classical quantity for initial stability.
- **C. Dupin (1814):** developed the *surface of buoyancy* and the *envelope of flotation*.

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Zindler Curves: the $\rho = \frac{1}{2}$ case

- **Zindler (1921):** discovered closed curves whose perimeter-halving chords all have the same length.
- **Auerbach (1938):** showed that such curves give planar floating bodies at density $\rho = \frac{1}{2}$.
- For Zindler curves, the halving chord is the geometric analogue of the waterline.



$$z(u) = x(u) + iy(u) = e^{2iu} + 2e^{-iu} + ae^{iu/2}, \quad u \in [0, 4\pi], \quad a \in \{4, 25\}.$$

Rigidity vs. Non-Rigidity

Perimetral density / case	Result	Authors
Ulam problem	is the sphere unique?	S. Ulam ($\sim$1935)
$\sigma = 0$	only the disk	L. Montejano (1974)
$\sigma = \frac{1}{2}$	infinitely many non-circular solutions	K. Zindler; H. Auerbach
$\sigma = \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{2}{5}$	only the disk	J. Bracho, L. Montejano, D. Oliveros (2004)

Generalizations

- **D. Ryabogin (2022):** gave a negative answer to Ulam's original 3D problem: there exists a strictly convex body of revolution in $\mathbb{R}^3$ of density $\frac{1}{2}$ that is non-spherical and still floats in equilibrium in every orientation.
- **D. I. Florentin, C. Schütt, E. M. Werner, and N. Zhang (2020/2022):** proved that among **origin-symmetric** convex bodies, density $\frac{1}{2}$ and flotation in every direction force the Euclidean ball in every dimension.
- **F. Wegner (2002/2007):** constructed explicit non-circular planar floating bodies for densities different from $\frac{1}{2}$.

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$\sigma = \frac{1}{6}$ with central symmetry	?	

Rigid Theorem for centrally symmetric Hexagons

Theorem (O.Asipchuk, M.K, P.Zatitskii)

The only centrally symmetric convex planar body that floats in equilibrium in every orientation with perimetral density $\sigma = \frac{1}{6}$ is the disk.

From Flotation to Polygons and Carousels

Theorem (H. Auerbach; J. Bracho, L. Montejano, D. Oliveros)

For $\sigma = \frac{1}{n}$, the following are equivalent:

1. *the planar floating-body problem with perimetral density σ;*
2. *the existence of a Zindler carousel with n chairs, i.e. a smooth one-parameter family of inscribed equilateral n-gons whose vertices cut the boundary into equal arcs.*

Carousel ODEs

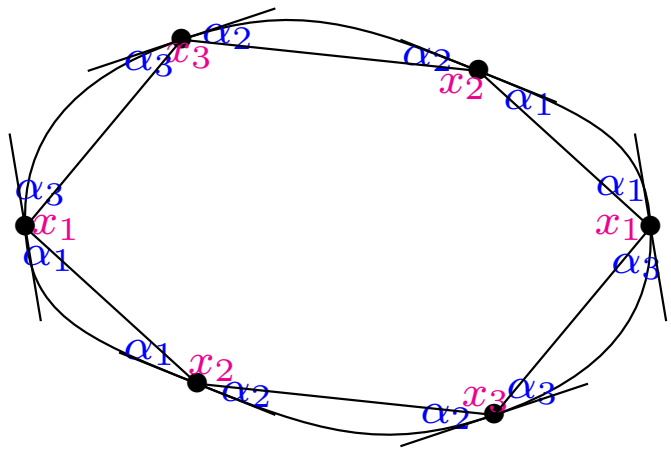


Figure: A central symmetric convex body K with an inscribed hexagon.

Proposition (J. Bracho, L. Montejano, D. Oliveros)

Let $x_i(t)$ be the interior angles and let $\alpha_i(t)$ be the auxiliary angles of an n-chair carousel. Then

$$\dot{x}_i(t) = \sin\big(\alpha_{i-1}(t)\big) - \sin\big(\alpha_i(t)\big).$$

Reduction to the Centrally Symmetric Hexagon

Proposition

For an equilateral centrally symmetric hexagon,

$$x_{i+3} = x_i, \qquad x_1 + x_2 + x_3 = 2\pi.$$

Setting

$$x = x_1(t), \qquad y = x_2(t), \qquad x_3 = 2\pi - x - y,$$

the carousel system reduces to

$$\begin{cases} \dot{x} = \cos(x+y) - \cos y, \\ \dot{y} = \cos x - \cos(x+y). \end{cases}$$

Strict convexity gives

$$\frac{\pi}{2} < x < \pi, \qquad \frac{\pi}{2} < y < \pi, \qquad x + y < \frac{3\pi}{2}.$$

Hamiltonian Structure

Lemma

The area S of the inscribed hexagon is preserved. With

$$H(x,y) = \frac{S}{4} = \sin x + \sin y - \sin(x+y),$$

the reduced system is Hamiltonian up to sign:

$$\dot{x} = -\frac{\partial \widetilde{H}}{\partial y}, \qquad \dot{y} = \frac{\partial \widetilde{H}}{\partial x}, \qquad \widetilde{H} = -H.$$

Range of the Hamiltonian

- H has a strict maximum at

$$\left(\frac{2\pi}{3}, \frac{2\pi}{3}\right), \qquad H_{\max} = \frac{3\sqrt{3}}{2}.$$

- On the boundary of the admissible region,

$$H \leq H_0 = 1 + \sqrt{2}.$$

- Therefore any nonconstant closed orbit relevant to a non-circular body must satisfy

$$1 + \sqrt{2} < H < \frac{3\sqrt{3}}{2}.$$

One-Dimensional Reduction and Period

Proposition

Introduce

$$u := \frac{x+y}{2}, \qquad v := \frac{x-y}{2}.$$

Then

$$\dot{u} = -\sin u \, \sin v.$$

Using conservation of H, eliminate v and obtain

$$\dot{u}^2 = \sin^2 u - \frac{(H + \sin 2u)^2}{4} =: Q_H(u).$$

If $u_-(H)$ and $u_+(H)$ are the turning points, the minimal period is

$$T(H) = 2 \int_{u_-(H)}^{u_+(H)} \frac{du}{\sqrt{Q_H(u)}}.$$

Period Bounds

Lemma

A quadratic comparison for Q_H on $[u_-, u_+]$ gives

$$\frac{4-\sqrt{2}}{2}(u-u_-)(u_+-u) \le Q_H(u) \le \frac{8}{3}(u-u_-)(u_+-u).$$

Hence

$$\pi\sqrt{\frac{3}{2}} < T(H) < \frac{2\pi}{\sqrt{(4-\sqrt{2})/2}} < 6.$$

Rotations and Periods

Consider the set of rotational symmetries

$$\mathrm{Rot}(K) := \{\theta \in [0, 2\pi) : \ R_\theta(K) = K\}.$$

If K is not a disk, $\mathrm{Rot}(K)$ is a finite cyclic group; define the *basic rotation angle*

$$\sigma_K := \min\Big(\mathrm{Rot}(K) \cap (0, 2\pi)\Big).$$

Correspondingly, define $\eta_K \in (0, P)$ to be the unique number such that

$$R_{\sigma_K}\Big(\omega(t)\Big) = \omega(t + \eta_K) \qquad \text{for all } t \in \mathbb{R}. \tag{1}$$

Let m be the order of $\mathrm{Rot}(K)$ (so $m\sigma_K = 2\pi$). Iterating (1) m times gives

$$m\,\eta_K = P.$$

In particular, since K is centrally symmetric, $R_\pi(K) = K$, thus $\pi \in \mathrm{Rot}(K)$ and therefore

$$\sigma_K = \frac{\pi}{k} \quad \text{for some } k \in \mathbb{N}, \qquad \text{and} \qquad \eta_K = \frac{P}{2k}. \tag{2}$$

Geometry and Period

One can show that

$$r^2 = 1 - 2H\cot u < (1+\sqrt{2})^2,$$

so $r(t)$ and $u(t)$ have the same minimal period $T(H)$, thus

$$r(t+\eta_K) = |\omega(t+\eta_K)| = |R_{\sigma_K}\omega(t)| = |\omega(t)| = r(t).$$

Therefore η_K is a period of $r(t)$. Since $T(H)$ is the minimal period of $r(t)$, there exists $m \in \mathbb{N}$ such that

$$\eta_K = m\,T(H).$$

combining with $\eta_K = \frac{P}{2k}$, it gives

$$\frac{P}{2k} = mT(H), \qquad k, m \in \mathbb{N}.$$

Contradiction and Conclusion

Recall

$$\pi\sqrt{\frac{3}{2}} < T(H) < \frac{2\pi}{\sqrt{(4-\sqrt{2})/2}} < 6. \tag{3}$$

on the other hand, using $12 \le P < 2\pi(1+\sqrt{2})$, we have

$$T(H) = \frac{P}{2km} \le \frac{2\pi(1+\sqrt{2})}{2km} = \frac{\pi(1+\sqrt{2})}{km},$$

which forces $km = 1$ and

$$T(H) = \frac{P}{2} \ge 6.$$

Thus we obtained two incompatible inequalities:

$$T(H) < 6 \qquad \text{and} \qquad T(H) \ge 6.$$

Further Directions

- What happens for the octagon cases $\sigma = \frac{1}{8}$ and $\sigma = \frac{1}{7}$?
- Can one prove monotonicity of the period $T(H)$ directly?
- How to use the Hamiltonian structure in an effective way?

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Thank you!

Any questions?